$=(A+B')\cdot((D'+(A'+C))\cdot(D'+B))$ (Theorem 4(a))

$=(A+B')\cdot(D'+(A'+C)\cdot B)$ (Postulate 4(b))

$=(A+B')\cdot D'+(A+B')\cdot((A'+C)\cdot B)$ (Postulate 4(a))

$=D'\cdot(A+B')+(A+B')\cdot(B\cdot(A'+C))$ (Postulate 3(b))

$=D'\cdot(A+B)+(A+B')\cdot(BA'+BC)$ (Postulate 4(a))

$=D'\cdot(A+B)+((A+B')\cdot BA'+(A+B')\cdot BC)$ (Postulate 4(a))

$=D'\cdot(A+B)+(BA'\cdot(A+B')+BC\cdot(A+B'))$ (Postulate 3(b))

$=D'\cdot(A+B)+((BA'\cdot A+BA'\cdot B')+(BCA+BC\cdot B'))$ (Postulate 4(a))

$=D'\cdot(A+B)+((BA'\cdot A+B'\cdot BA')+(BCA+B'\cdot BC))$ (Postulate 3(b))

$=D'\cdot(A+B)+((B\cdot(A'A)+(B'B)\cdot A')+(BCA+(B'\cdot B)\cdot C))$ (Theorem 4(b))

$=D'\cdot(A+B)+((B\cdot(AA')+(BB')\cdot A')+(BCA+(BB')\cdot C))$ (Postulate 3(b))

$=D'\cdot(A+B)+((B\cdot 0+0\cdot A')+(BCA+0\cdot C))$ (Postulate 5(b))

$=D'\cdot(A+B)+((B\cdot 0+A'\cdot 0)+(BCA+C\cdot 0))$ (Postulate 3(b))

$=D'\cdot(A+B)+((0+0)+(BCA+0))$ (Theorem 2(b))

$=D'\cdot(A+B)+(0+BCA)$ (Postulate 2(a))

$=D'\cdot(A+B)+(BCA+0)$ (Postulate 3(a))

$=D'\cdot(A+B)+BCA$ (Postulate 2(a))

c) $[(AB)'A][(AB)'B]$

$=[(A'+B')A][(A'+B')B]$ (Theorem 5(b))

$=[A(A'+B')][B(A'+B')]$ (Postulate 3(b))

$=[AA'+AB'][BA'+BB']$ (Postulate 4(a))

$=[0+AB'][BA'+0]$ (Theorem 2(b))

$=[AB'+0][BA'+0]$ (Postulate 3(a))

$=(AB')\cdot(BA')$ (Postulate 2(a))

$=A\cdot(B'\cdot(BA'))$ (Theorem 4(b))

$=A\cdot((B'B)\cdot A')$ (Theorem 4(b))

$=A\cdot((BB')\cdot A')$ (Postulate 3(b))

$=A\cdot(0\cdot A')$ (Theorem 2(b))

$=A\cdot(A'\cdot 0)$ (Postulate 3(b))

$=A\cdot 0$ (Theorem 2(b)) $=0$ (Theorem 2(b)). The complement of 0 is 1.